

Purplle: Creating hyper- personalized beauty experiences with cloud- based data

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ABOUT PURPILLE

Purplle deepened its integration with Google Cloud to help retailers handle more data for personalization for millions of customers.

Google
Cloud
results

Price
automation



About Purplle

Founded in 2012, [purplle.com](https://www.purplle.com/)

(<https://www.purplle.com/>) is one of India's largest e-beauty destinations with 7 million monthly active users. Purplle has over 1000+ brands with over 60,000 products, available on its website and app. The company's investors include Kedaara, Premji Invest, Sequoia Capital India, Goldman Sachs, Verlinvest, Blume Ventures, and Paramark Ventures.

Purplle aims to deliver a personalized retail experience with data and averages eight million

- Increases company's daily revenue by up to 20% with improved pricing accuracy and automation
- Cuts data processing time by 60% with data capture shift from ETL to ELT
- Serves 8 million personalized requests per day under 40-60 milliseconds
- Easily handles a surge of 24 million personalized requests

goes 1
to live
weeks

personalization requests on its app. By adopting cloud technology, Purplle has improved the experience of its rapidly growing customer base. In June 2022, the company achieved unicorn status after raising US\$33 million in its Series E round. With over 2400 employees and growth of >90% GMV CAGR for the last 3 years, Purplle is well-positioned to be a leader in the burgeoning Indian e-beauty space.

Industries: Retail & Consumer Goods

Location: India

(<https://cloud.google.com/automl/>), [BigQuery](#).

About MediaAgility

Based in the U.S, MediaAgility is a premier global digital consultancy that helps companies apply new technologies to build scalable cloud-based infrastructure, to enhance business agility for real customer impact.

(<https://cloud.google.com/compute/>),

Container Engine

Google Cloud (<https://cloud.google.com/>),
(<https://cloud.google.com/aiplatform/docs/quickstart>),

Cloud AutoML (<https://cloud.google.com/automl/>)

Cloud Dataproc

BigQuery (<https://cloud.google.com/bigquery/>)

Kubernetes Engine

Certificate Authority Service

(<https://cloud.google.com/certificate-authority-service/>)

(<https://cloud.google.com/pubsub/>)

Cloud Bigtable (<https://cloud.google.com/bigtable/>)

With thousands of beauty and s
the market, it can be difficult for people to choose

the right products for themselves to solve the challenge that **Purple** (https://www.purple.com) of India's largest beauty ecomm to solve by using technology to improve customer experience. The fast-growing company achieved its unicorn status in 2019 with sights on going beyond delivering products to millions of monthly

Cloud Composer

(https://cloud.google.com/composer/)

Cloud Functions

(https://cloud.google.com/functions/)

Cloud Storage (https://cloud.google.com/storage/)

Compute Engine

(https://cloud.google.com/compute/)

Container Registry

(https://cloud.google.com/container-registry/)

Dataflow (https://cloud.google.com/dataflow/)

Cloud Dataproc

(https://cloud.google.com/dataproc/)

Kubernetes Engine

(https://cloud.google.com/kubernetes-engine/)

Cloud Pub/Sub (https://cloud.google.com/pubsub/)

From helping customers identify beauty trends, to trying makeup virtually, Purple is looking to improve its recommendation capabilities, and take its omnichannel strategy to India's massively underpenetrated beauty retail market. To support its mission, the company embarked on a journey to migrate its infrastructure (https://cloud.google.com/) in 2019.



Expanding the possibilities for personalization

The key to creating a repeat and growing customer base is to implement personalization from the start and to do it with more automation. Purple's app takes up to 40 different user data points like their

skin type, facial structure, ethnicity, and scalp condition and returns suggestions on the right product for their personal needs over the short- or long-term.

"Technology plays a vital role in making our data model successful. It helps us go beyond just being a marketplace and utilizing all the information we get from our customers to help them come back to our platform," explains Vivek Parihar, Head of Engineering at Purplle.

Behind the scenes, Purplle's data science team runs hundreds of scripts to train various recommendations and store models with Cloud Bigtable (<https://cloud.google.com/bigtable>). The fully managed, scalable NoSQL database gives their team the flexibility to store data in a columnar format, and its versioning capabilities mean that Purplle's team doesn't have to worry about failed scripts or garbage collection, which Cloud Bigtable automatically handles.

Regardless of whether Purplle needs to serve eight million personalized requests per day or 24 million requests at its peak, Bigtable's recommendation P99 latency, which is below 60 milliseconds, ensures that their customers are served quickly.

Dataproc is also used by Purplle's data science team to structure its data streaming and transformation from its data lake on Cloud Storage (<https://cloud.google.com/storage>) to its data warehouse on BigQuery (<https://cloud.google.com/bigquery>). Structured data is

used in several of its systems, including the presentation, analytics, and recommendation systems that deliver the most appropriate products to its customers.

"When we look at the role Dataproc (<https://cloud.google.com/dataproc>) plays, it has really helped us train models involving hundreds of gigabytes of data, and Dataproc gives us good auto-scaling capabilities to pre-process and train our models on the fly," adds Parihar.

This level of personalization keeps customers delighted with meaningful, value-added suggestions, and has helped maintain the company's strong conversion rate, Parihar says. Both benefits are especially crucial for the company's small team, which has relied on technology to multiply the effectiveness of each team member.

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Handling exponential data growth while keeping costs low

The team uses Dataflow

(<https://cloud.google.com/dataflow>) for data analytics,

Dataproc for data ingestion, and Pub/Sub

(<https://cloud.google.com/pubsub>) for event streaming.

Because of the easy integration of these tools into the larger infrastructure, engineers can research, develop, and launch new customer experiences without worrying about cost-per-hour as they typically would, since these managed services only register a cost if and when they are used. This helps Purplle keep costs low while building new products to move faster in the market.

Another key element of Purplle's research and development is Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine>), which they use to quickly create pods for more intensive model training that requires very high RAM and CPU resources. Purpille can run between 100 to 150 pods daily to train its model in a specified timeframe. This happens automatically without them worrying about human resource allocation, which is important because the retailer doesn't have a fully managed DevOps team. Yet, they can cost-effectively train their models by tasking one or two people and have their effectiveness multiplied by technology.

"During a sale, we would see a huge spike in our component requests, which would normally be problematic due to high traffic volume. Google Kubernetes Engine enables us to autoscale the deployment of pods the moment we see a spike. For example, we can increase our pod size by ten times in under a minute," Parihar says.

Keeping its products competitively priced is another area that Purpille needs to keep in check if they want to ensure customer loyalty. "Pricing is a sensitive aspect for any ecommerce platform because incorrect pricing can cost you a lot of money, particularly when multiplied by thousands of orders," Parihar describes.

The team turned to Cloud AutoML for pricing automation to solve this issue. Because AutoML enables developers with limited machine learning experience to build a custom machine learning model in minutes, they could task just one data analyst to run the project.

AutoML demonstrated that Purple could automate pricing and realize 15% to 20% in revenue savings daily. That proof-of-concept project went live within two weeks, and the new pricing mechanism updates every four hours automatically with Cloud Functions (<https://cloud.google.com/functions>).

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Creating an omnichannel experience with real-time data

While the company's core strength and biggest revenue source is its online channels, taking Purple to the next level will require more powerful and smarter tools to help it do more with its data.

In its earlier years, Purple used Extract, Transform, Load (ETL) for its change data capture (CDC) process but this became too slow as its data lake grew. "We figured that with all the managed services at our disposal, from BigQuery to Bigtable, Dataproc and Pub/Sub, we could shift from ETL to ELT (Extract, Load, Transform) to speed up data processing," says Parihar. This switch provided an efficiency gain of 60% because of the reduction in time and bandwidth.

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When real-time data becomes an advantage for Purplle, it can venture into complex, new scenarios that help the retailer use data to realize even more ambitious customer experiences. "With the latest round of funding, we have a very different set of expectations with our technology as we venture into offline channels," says Parihar.

The new venture is one the company is taking cautiously. "It's a new ballgame. You need to merge the user experience as they transition from offline to online, or online to offline. The challenge is to create seamless omnichannel experiences for our customers," he adds.

One exciting way Purplle is looking to grow its omnichannel offering is to replace the need for in-store beauty advisors with a more convenient and intelligent version on tablets and other mobile devices that can offer recommendations based on customer photos and preexisting customer data.

But it's not getting that done alone. "More than being told which tools to use, the Google Cloud team

understands our business, and helps us with the integration of new tools that can support Purple's growth," Parihar says. "We look forward to leveraging more Google Cloud products and tools as we continue to scale our business."